

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

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REG NO: 192372098

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS:

Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

*I **G.CHAKRADHAR(192311439)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

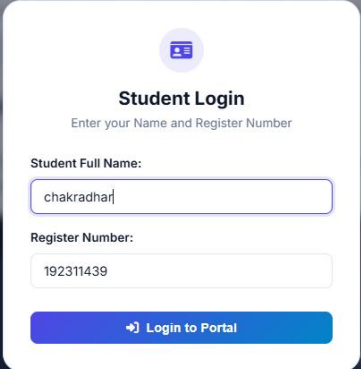
Signature of the student: E.HEMANTH REDDY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

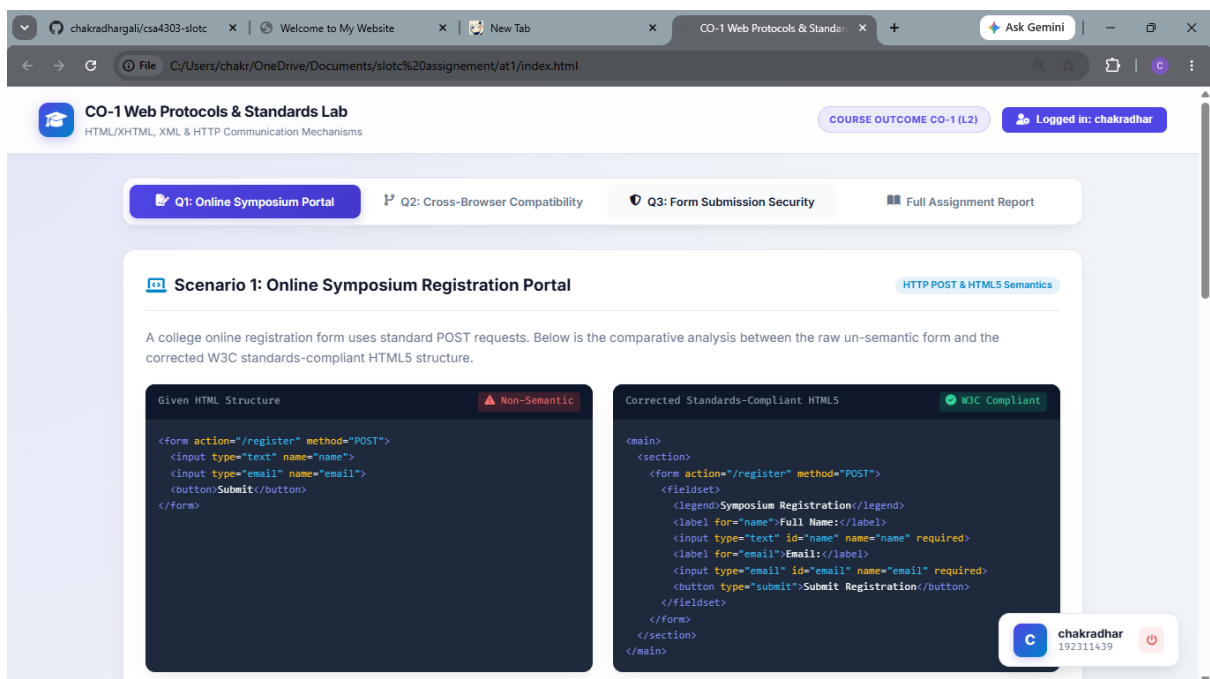
OUTPUT:

1.login:



The image shows a 'Student Login' form overlay on a blurred background of a web page. The form has a title 'Student Login' with a subtitle 'Enter your Name and Register Number'. It contains two input fields: 'Student Full Name:' with the value 'chakradhar' and 'Register Number:' with the value '192311439'. A blue button labeled 'Login to Portal' is at the bottom.

2.question-1:



The screenshot shows a web browser with a URL bar displaying 'C:/Users/chakr/OneDrive/Documents/slot%20assignment/at1/index.html'. The page title is 'CO-1 Web Protocols & Standards Lab' with a subtitle 'HTML/XHTML, XML & HTTP Communication Mechanisms'. A navigation bar includes 'Q1: Online Symposium Portal', 'Q2: Cross-Browser Compatibility', 'Q3: Form Submission Security', and 'Full Assignment Report'. A 'Logged in: chakradhar' button is visible. The main content area is titled 'Scenario 1: Online Symposium Registration Portal' with a sub-header 'HTTP POST & HTML5 Semantics'. The text describes a college online registration form using standard POST requests. Below this, two code blocks are shown: 'Given HTML Structure' (Non-Semantic) and 'Corrected Standards-Compliant HTML5' (W3C Compliant). The 'Given HTML Structure' code is:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

 The 'Corrected Standards-Compliant HTML5' code is:

```
<main>
  <section>
    <form action="/register" method="POST">
      <fieldset>
        <legend>Symposium Registration</legend>
        <label for="name">Full Name:</label>
        <input type="text" id="name" name="name" required>
        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>
        <button type="submit">Submit Registration</button>
      </fieldset>
    </form>
  </section>
</main>
```

 A user profile card for 'chakradhar' with ID '192311439' is visible in the bottom right corner.

CO-1 Web Protocols & Standards Lab
HTML/XHTML, XML & HTTP Communication Mechanisms

COURSE OUTCOME CO-1 (L2)

Logged in: chakradhar

Live HTTP Request & Response Cycle Simulator

Interactive Test Bench

1TCP & TLS Handshake

2Send HTTP POST

3Server Validation

4HTTP 200 / 302 Response

Submit Registration Form

Student Full Name:
Alex Rivera

Student Email Address:
alex.rivera@university.edu

Trigger HTTP POST Request

HTTP Packet & Headers Inspector
// Click 'Trigger HTTP POST Request' to view raw HTTP headers and response cycle ...

Analytical Answers

1. Missing Semantic Elements
The raw snippet lacks <main>, <section>, <fieldset>, <legend>, and <label> tags. Inputs are floating without explicit labeling, preventing screen readers from linking input purpose to blind users.

2. Standards Compliance

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Question2:

CO-1 Web Protocols & Standards Lab
HTML/XHTML, XML & HTTP Communication Mechanisms

COURSE OUTCOME CO-1 (L2)

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Scenario 2: Cross-Browser Compatibility & Parsing

Quirks Mode & Unclosed Tags

Missing <!DOCTYPE html> forces modern browser rendering engines into Quirks Mode while unclosed tags trigger browser DOM auto-correction heuristics.

Invalid HTML Snippet
Syntax Errors

W3C Compliant Corrected HTML
Standards Mode

Analytical Answers

1. Nesting & Closing Tag Errors
Missing <!DOCTYPE html>, line 1: Unclosed <h1> before <div>, line 1: Unclosed <div> before </div>

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≡ Analytical Answers

1. Nesting & Closing Tag Errors

Missing `<!DOCTYPE html>` line 1. Unclosed `<h1>` before `<p>` tag. Unclosed `<p>` tag before `</div>`.

2. Browser Interpretation & Quirks Mode

Without `<!DOCTYPE html>`, browser triggers **Quirks Mode**, emulating legacy IE5/6 box models and font rules. Chrome auto-closes `<h1>` upon reading `<p>`, whereas non-standard engines nest `<p>` inside `<h1>`, breaking layout across platforms.

3. Missing Semantic Landmarks

Lacks `<header>`, `<main>`, `<section>`, and essential meta tags (`charset="UTF-8"`, responsive viewport).

4. Techniques for Cross-Browser Compatibility

1. Enforce Standards Mode with `<!DOCTYPE html>`.
2. Run W3C Markup Validation.
3. Apply Modern CSS Normalize/Reset.
4. Autoprefix CSS vendor extensions.
5. Perform automated multi-browser engine testing (Playwright/Blink/WebKit).

Question 3:

CO-1 Web Protocols & Standards Lab
HTML/XHTML, XML & HTTP Communication Mechanisms

COURSE OUTCOME CO-1 (L2)

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Scenario 3: Form Submission Failure & HTTP Security

Critical Form Defect

Analyzing why `<button type="button">` prevents form dispatch and why `method="get"` exposes credentials in cleartext URLs.

Defective Form Code

Insecure & Non-Functional

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <!-- Defect: type="button" does NOT submit form -->
  <button type="button">Login</button>
</form>
```

Correct & Secure Form Implementation

Secure & Functional

```
<form action="https://example.com/api/login" method="POST">
  <input type="hidden" name="csrf_token" value="xyz123">

  <label for="user">Username:</label>
  <input type="text" id="user" name="username" required>

  <label for="pass">Password:</label>
  <input type="password" id="pass" name="password" required>

  <!-- Fixed: type="submit" triggers HTTP POST -->
  <button type="submit">Login</button>
</form>
```

Live Form Submission & Security Simulator

Test Button Behavior & GET Exposure

Test Bench Controls

Test Username:
admin_user

Test Password:

Simulated Browser URL Bar & Network Inspector

https://website.com/login

// Click a test button above to simulate browser form submission behavior ...

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CO-1 Web Protocols & Standards Lab
 HTML/XHTML, XML & HTTP Communication Mechanisms

COURSE OUTCOME CO-1 (L2)

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Live Form Submission & Security Simulator

Test Bench Controls

Test Username:

Test Password:

▶ Test Broken Form (type="button")

▶ Test Secure Form (type="submit" & POST)

Simulated Browser URL Bar & Network Inspector

https://website.com/login

// Click a test button above to simulate browser form submission behavior ...

Analytical Answers

1. Identification of Form Defect

- <button type="button"> does NOT trigger form submission.
- method="GET" sends passwords in plain text in the URL string.

2. Why HTTP Request is Not Triggered

A button configured with type="button" only dispatches a generic JavaScript click event. It does **not** fire the DOM submit event on the parent <form> element. As a result, the browser never constructs or sends an HTTP request.

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Full assignment report:

CO-1 Web Protocols & Standards Lab
 HTML/XHTML, XML & HTTP Communication Mechanisms

COURSE OUTCOME CO-1 (L2)

Logged in: chakradhar

Q1: Online Symposium Portal
 Q2: Cross-Browser Compatibility
 Q3: Form Submission Security
 Full Assignment Report

Complete Course Outcome Assignment Document

Print / Save Report

Course Outcome CO-1 Solutions

Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Summary Table of Key Concepts Covered

Topic	Identified Problem	Correct Standard Solution
Q1: Semantics	Floating inputs lacking <label> & landmarks	Wrap in <main>, <fieldset>, and <label for="...">
Q2: Parsing	Missing <!DOCTYPE> (Quirks Mode) & unclosed tags	Include <!DOCTYPE html> & explicit closing tags
Q3: HTTP Security	type="button" fails dispatch; GET leaks password	Use type="submit", POST method & HTTPS encryption

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Result:

We have successfully Developed well-structured, standards-compliant web pages using HTML/XHTML and XML, and we applied Internet protocols and HTTP communication mechanisms for web content delivery